

The Number System: A Comprehensive Overview

Mathematical Foundations

Prepared by Nag's Academy

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INTRODUCTION

The number system is a mathematical framework used to express numbers and represent the magnitude of quantities. From the basic counting numbers used by early civilizations to the complex numbers used in modern engineering, the evolution of the number system is fundamental to our understanding of the universe.

CLASSIFICATION OF NUMBERS

Numbers are categorized into different sets based on their properties.

2.1 Natural and Whole Numbers

Natural numbers ($\mathbb{N}$) are the basic counting numbers: $\{1, 2, 3, \dots\}$. When zero is included, they become Whole numbers: $\{0, 1, 2, \dots\}$.

2.2 Integers ($\mathbb{Z}$)

Integers include all whole numbers and their negative counterparts:

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

2.3 Rational and Irrational Numbers

- **Rational Numbers ($\mathbb{Q}$):** Any number that can be expressed as a fraction $\frac{p}{q}$ where $p, q \in \mathbb{Z}$ and $q \neq 0$.
- **Irrational Numbers ($\mathbb{I}$):** Numbers that cannot be expressed as simple fractions, such as π , e , or $\sqrt{2}$. Their decimal expansions are non-terminating and non-recurring.

2.4 Real and Complex Numbers

The set of Real numbers ($\mathbb{R}$) is the union of rational and irrational numbers. Beyond this lies the Complex numbers ($\mathbb{C}$), expressed as:

$$z = a + bi$$

where a and b are real numbers, and $i = \sqrt{-1}$.

POSITIONAL NOTATION AND BASES

A number system can be defined by its base (radix). The most common systems used in computing and mathematics are listed below.

Table 1: Common Number Bases		
System	Base	Digits Used
Binary	2	0, 1
Octal	8	0–7
Decimal	10	0–9
Hexadecimal	16	0–9, A–F

PROPERTIES OF NUMBER SYSTEMS

Operations within the number system typically follow specific laws, assuming the numbers belong to a field like $\mathbb{R}$:

1. **Commutative Law:** $a + b = b + a$ and $ab = ba$.
2. **Associative Law:** $(a + b) + c = a + (b + c)$.
3. **Distributive Law:** $a(b + c) = ab + ac$.

CONCLUSION

Understanding the hierarchy and properties of the number system is essential for higher mathematics and science. Each set of numbers builds upon the previous one, expanding the possibilities of calculation and theoretical exploration.